Class 18 van der Pol oscillator

- No class Friday.
- There is a pre-class assignment for the Monday after spring break.
- Quiz 02 is scheduled for Monday Mar 30

Big picture

We have been working on how to construct a phase portrait for a 2d system in $\mathbb{R}^2$., and have specifically been working to determine whether or where a phase portrait might have closed trajectories.

Today we will work with the van der Pol oscillator, which is a specific example system that has oscillatory behavior and shows timescale separation.

Liapunov functions:
A **Liapunov function** for a system is a continuously differentiable function $V(x, y)$ with the following properties:

1. $V(x, y) > 0$ for all (x, y) not equilibria,
2. If (x, y) is an equilibrium point, then $V(x, y) = 0$,
3. $\dot{V} < 0$ along all trajectories to an equilibrium.

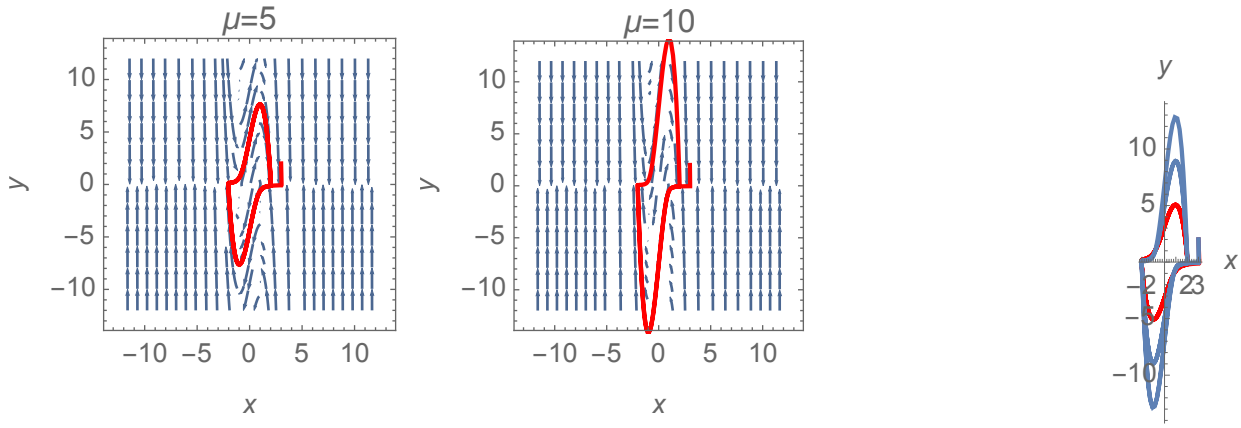
The existence of a **Lyapunov function** on a region U allows us to **rule out closed trajectories** in that region.

Theorem: If L is a Liapunov function for an equilibrium $\mathbf{x}$ then the equilibrium is **asymptotically stable**: every point in U will approach $\mathbf{x}^*$ as $t \to \infty$.

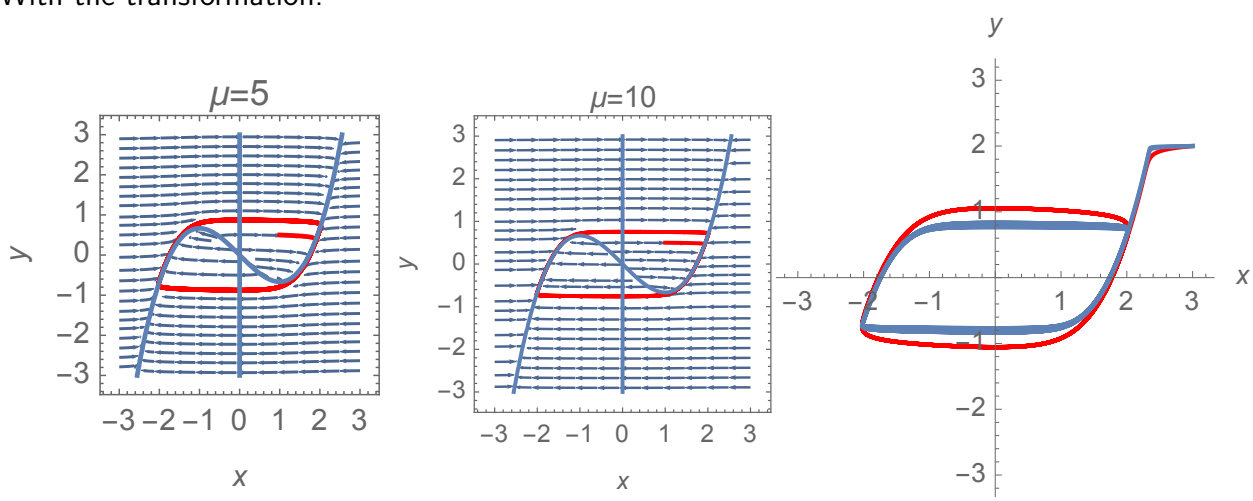
Bendixson's criterion rules out closed trajectories on some regions of a plane.
On regions where $f_x + g_y$ has a single sign, a closed trajectory is not possible.
Justification: The **flux** of a vector field across a closed trajectory is zero. On a closed trajectory, the vector field is tangent to the trajectory: $\mathbf{F} \cdot \hat{\mathbf{n}} = 0$ everywhere on the curve.
By the **divergence form of Green's theorem**, $\displaystyle\oint_{\partial R} \mathbf{F} \cdot \hat{\mathbf{n}}\, ds = \iint_R \nabla \cdot \mathbf{F}\, dA$ where R is a region and $C = \partial R$ is its boundary curve. The integral of the divergence over a region is equal to the flux out the boundary of the region. If $\iint_R \nabla \cdot \mathbf{F}\, dA \neq 0$ then $\displaystyle\oint_{\partial R} \mathbf{F} \cdot \hat{\mathbf{n}}\, ds \neq 0$ and the boundary of R is not a closed trajectory.

Van der Pol

Without a Lienard transformation:



With the transformation:



Activity

Teams 3 & 4, post photos of your work to the class Google Drive (see Canvas for link). Make a folder for today's class if one doesn't exist yet.

1. (convince yourself of the details: van der Pol example) After a change of variables the van der Pol system is

$$\dot{x} = \mu(y - \frac{1}{3}x^3 + x)$$
$$\dot{y} = -\frac{1}{\mu}x.$$

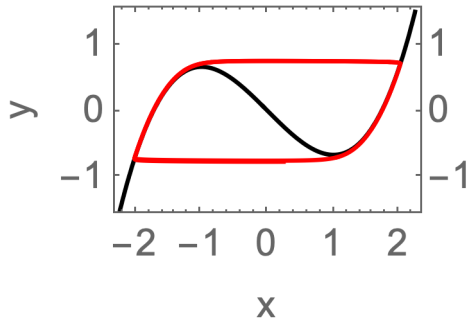
Consider the case where $\mu \gg 1$.

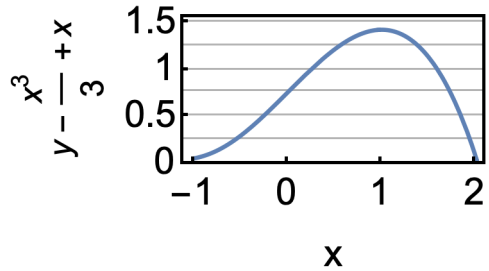
(a) In this relaxation oscillation, the trajectory is moving quickly when it jumps between the two parts of the $\dot{x} = 0$ nullcline.

In the plot on the left, the red curve is the trajectory.

In the plot on the right, the blue curve is showing $y - \frac{1}{3}x^3 + x$ as a function of x along the upper jump, where $y \approx 2/3$, $1 \leq x \leq 2$.

By looking at the values of $y - \frac{1}{3}x^3 + x$ in the graph, convince yourself that $\dot{x} > 0.1\mu$ for most of the jump.





We say this velocity is of the order of μ, or is $\mathcal{O}(\mu)$, because it is bounded above by a constant multiple of μ.

(b) The trajectory is traversing a distance of about 3 as it jumps. Use the nullcline to find the start and end point of the jump to confirm the distance.

$\langle \dot{x} \rangle \approx \mu/3$ along the jump.

Combine the approximate distance and the approximate average velocity to find the μ-dependence of the time that a particle spends jumping.

(c) While moving along the nullcline, the trajectory moves about 1 unit in x and a bit less than 2 units in y. It is basically moving along the curve $y = \frac{1}{3}x^3 - x$ (it is not quite on the curve, but it is close to that curve the whole time). The time it spends traversing the curve is a constant multiple of μ^k for some integer k.

To estimate time (just as we did for the oscillators in chapter 4), we set up an integral of the form $\displaystyle\int_{x_1}^{x_2} \frac{dt}{dx} dx$ or something like this. This integral shows us how the time depends on μ. Using

$$\int_{x_1}^{x_2} \frac{dt}{dx} dx$$

doesn't work so well. It puts $y - (\frac{1}{3}x^3 - x)$ in the denominator (so a dependence on x and y, not just x). What is wrong with having a dependence on x and on y in the integral?

(d) We could try again with

$$\int_{y_1}^{y_2} \frac{dt}{dy} dy.$$

Argue that this leads to a problem, too, and isn't something we can integrate.

(e) So actually, Steve used

$$\int_{x_1}^{x_2} \frac{dt}{dy}\frac{dy}{dx} dx.$$

(This is an example of persisting until something works, and luckily getting something to work before we run out of options). Confirm that this expression results in something that is integrable.

Note that for $\dfrac{dy}{dx}$ we're thinking of trajectories as, to good approximation, being stuck on the nullcline, so compute this by assuming the trajectory is exactly on the nullcline.

(f) Use the setup of this final integral to identify k. There is no need to evaluate the integral. We just want to learn the μ dependence of the time.

(g) Compare the amount of time spent jumping to the amount of time spend moving along the curve. (We have the timescales of these processes, and not the exact amounts of time, so compare the timescales).

Answers:

(a) When the trajectory is flowing across, $\dot{y}$ is small: $|x| < 3$ or so, and μ is big, so $|\frac{x}{\mu}| < \frac{3}{\mu}$, which is small (specifically order of $\frac{1}{\mu}$. This means the motion is basically horizontal. And it is moving at a speed of $\mu(y - x^3/3 + x)$ in the horizontal direction. Away from the nullcline itself, $y - x^3/3 + x$ is between 0.1 and 1.4 for almost the whole distance across, so $\dot{x}$ will be between 0.1μ and 1.4μ for most of the jump.

(b) It jumps across a distance of maybe 3 as it moves horizontally. Since $distance/time = velocity$ the time is $distance/velocity$, so it is proportional to $\frac{1}{\mu}$. This will be very small and means that it jumps across pretty quickly.

(c) Along the nullcline, x and y each depend on t, but they also have a relationship with each other. The integral $\int_{x_1}^{x_2} \frac{1}{y+x-x^3/3} dx$ isn't something that we can integrate because y is coupled to x as we move along the nullcline, so we can't treat y as a constant while we integrate with respect to x. Instead, we would need to write y in terms of x.

(d) This one is $\int_{y_1}^{y_2} -\frac{\mu}{x} dy$ so also mixes together x and y. As y changes, x also changes, so we can't integrate this without knowing how they relate to each other. Instead, we would need to write x in terms of y.

(e) On the nullcline, $y = -x + x^3/3$ so $\frac{dy}{dx} = -1 + x^2$. $\displaystyle\int_{x_1}^{x_2} -\frac{\mu}{x}(x^2 - 1)dx$. This is an integral where the integrand doesn't depend on y or t but just on x, and it is being integrated dx, so this one can be computed as written.

(f) $\frac{dt}{dy} = -\mu\frac{1}{x}$ and $\frac{dy}{dx} = x^2 - 1$ on the nullcline. So $T = \displaystyle\int_{x_1}^{x_2} -\mu\frac{1}{x}(x^2 - 1)dx = \mu\int_{x_1}^{x_2} -\frac{1}{x}(x^2 - 1)dx$. This is μ multiplied by a constant that doesn't depend on μ. So this time is proportional to μ meaning that $k = 1$.

(g) The jump is order of $\frac{1}{\mu}$ and the motion along the curve is order of μ. This means we spend a ton more time on the curve compared to doing the jump.

2. Set up an integral in a single variable for the amount of time it takes a trajectory to traverse the curve $y = F(x)$ from $x = x_0$ to $x = x_1$, when $\dot{y}$ is given by $\dot{y} = g(x) = x$ and $F(x) = x^2 - 2x$

Answer: $T = \displaystyle\int_{x_0}^{x_1} \frac{2x-2}{x} dx$

Explanation: Using the chain rule, $T = \displaystyle\int_{x_0}^{x_1} \frac{dt}{dy}\frac{dy}{dx} dx$. Notice that x is the variable of integration and the bounds are given as x values. The dimension of the integrand is time.

To find $\frac{dy}{dx}$: $y = F(x)$ on our trajectory so we have $\frac{dy}{dx} = F'(x) = 2x - 2$ on our trajectory.

To find $\frac{dt}{dy}$: Using $\frac{dt}{dy} = 1/\dot{y}$, we have $T = \displaystyle\int_{x_0}^{x_1} \frac{1}{x} F'(x)dx$.

Extra vocabulary / extra facts:

- A dynamical system is called **weakly nonlinear** when the system is a small perturbation away from a linear system.
- In its weakly nonlinear form, the van der Pol system can be written as $\ddot{x}+x+\epsilon\dot{x}(x^2-1) = 0$ where ϵ is a small parameter.

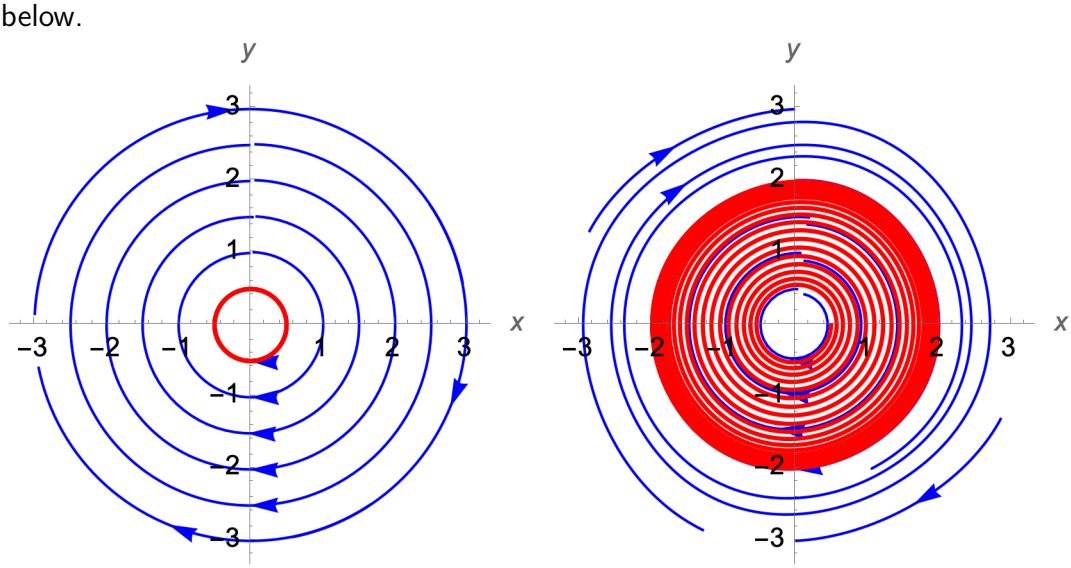
  This is a **small perturbation** from the system $\ddot{x}+x = 0$, which is a conservative system ($E(x,\dot{x}) = \frac{1}{2}x^2+\frac{1}{2}\dot{x}^2$ is conserved) and has solutions of the form $x(t) = A\cos t+B\sin t$ (for arbitrary A and B).
- The presence of the $\epsilon\dot{x}(x^2-1)$ term changes the vector field only very slightly from the vector field in the system $\ddot{x}+x = 0$.
- We can use an **energy method** for approximating the location of a limit cycle in the weakly nonlinear system. On a limit cycle, we move periodically in phase space, so $E(x,\dot{x})$ will be periodic. Look at ΔE over one cycle. This will be 0 on a limit cycle, so to approximate the limit cycle, work to find a trajectory over which $\Delta E = 0$.

3. (weakly nonlinear van der Pol)

   Let $E(x,\dot{x}) = \dfrac{1}{2}x^2 + \dfrac{1}{2}\dot{x}^2$ or $E(x,y) = \dfrac{1}{2}x^2 + \dfrac{1}{2}y^2$

   (a) Find $\dfrac{dE}{dt}$ for the weakly nonlinear van der Pol oscillator, where $x + \ddot{x} = -\epsilon\dot{x}(x^2-1)$.
   Write $\dot{E}$ in terms of just x and $\dot{x}$ or y.

   (b) Along a trajectory, we have $\Delta E = \displaystyle\int_0^T \frac{dE}{dt}dt$. The weakly nonlinear van der Pol is very close to the system $\ddot{x}+x = 0$. Two trajectories are shown for each system in the plots below.



   In the $\ddot{x}+x = 0$ system, the period of cycles is $T = 2\pi$, and $x(t) = A\cos t$ is a solution for any A.

In the weakly nonlinear van der Pol system, assume there exists a limit cycle (we know that there is one from Lienard's theorem), and that it is of the form $x(t) = A\cos t$ with period $T \approx 2\pi$. Find the value of A associated with the limit cycle via the following steps:

- Assume $x(t) = A\cos t$ on the limit cycle, and use this $x(t)$ to find $\dot{x}$.
- Substitute $\dot{x}$ and x into $\Delta E \approx -\epsilon \int_0^{2\pi} \dot{x}^2(x^2-1)dt$
- Using $\frac{1}{2\pi}\int_0^{2\pi} \sin^2 t\ dt = \frac{1}{2}$ and $\frac{1}{2\pi}\int_0^{2\pi} \cos^2 t \sin^2 t\ dt = \frac{1}{8}$, find a value of A such that $\Delta E = 0$.
- Check the A you found against what is happening in the phase portrait above.

Answers:

1. $\dot{E} = x\dot{x} + \dot{x}\ddot{x} = \dot{x}(x + \ddot{x}) = -\epsilon\dot{x}^2(x^2-1)$

2. $\dot{x} = -A\sin t$. $\Delta E = -\epsilon\int_0^{2\pi} A^2\sin^2 t(A^2\cos^2 t - 1)dt = -\epsilon\,(A^4/8 - A^2/2)$. $A^4/8 - A^2/2 = 0$ so $A^2/4 = 1 \Rightarrow A = 2$. This looks like it matches the location where the red trajectory saturates.

4. (Ruling out closed orbits) Let

$$\begin{aligned}\dot{x} &= y\\ \dot{y} &= x - x^3 - \mu y, \quad \mu \geq 0.\end{aligned}$$

This system is known as the unforced Duffing oscillator. (Duffing published work on this kind of equation in 1918, and it became a popular model in the 1970s). The μy term is a damping term.

(a) This system can also be written as $\ddot{x} = x - x^3 - muy$. How does this differ from the similar form that we have seen for a conservative system?

(b) Use Bendixson's criterion to show that the system has no closed orbits for $\mu > 0$.

Answers:

a: The conservative version is $\dot{x} = y$, $\dot{y} = F(x)$ but this is $\dot{y} = g(x, y)$ (there is a y dependence in the $\dot{y}$ equation).

b: The vector field is $\vec{F} = y\vec{i} + (x - x^3 - \mu y)\vec{j}$. The divergence is $0 - \mu$ so the divergence is negative everywhere so long as $\mu > 0$. That means the integral of the divergence over any region with nonzero area will be negative, so $\oint_C \vec{F}\cdot n\ ds$ is negative. It cannot be zero, so no closed trajectories.

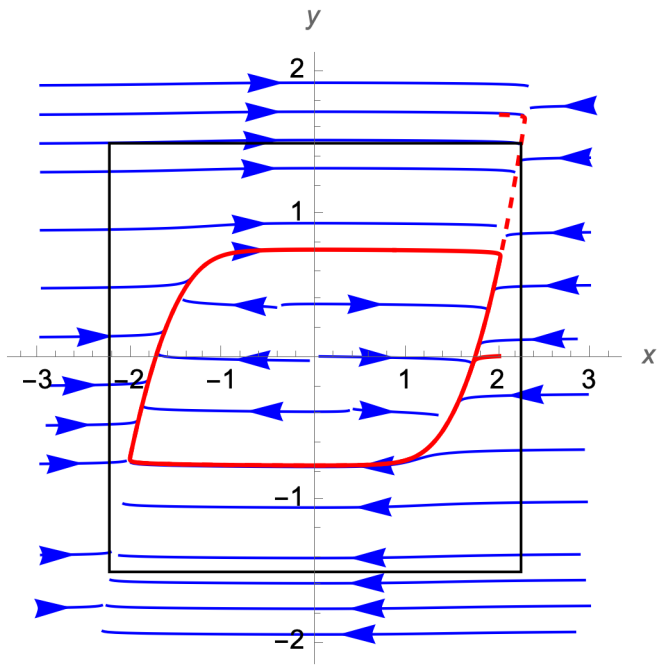
5. Find a restriction on a and b such that $V(x, y) = ax^2 + by^2$ is a Liapunov function for

$$\begin{aligned}\dot{x} &= y - x^3\\ \dot{y} &= -x - y^3\end{aligned}$$

Answers:

$V(x, y) > 0$ away from the origin for $a > 0, b > 0$. $\dot{x} = 0, \dot{y} = 0$ has a fixed point at the origin. $\dot{V} = 2ax\dot{x} + 2by\dot{y} = 2ax(y - x^3) + 2by(-x - y^3) = 2axy - 2ax^4 - 2bxy - 2by^4$. Need $a = b$ and then we have $-2ax^4 - 2by^4$ which is negative away from the origin.

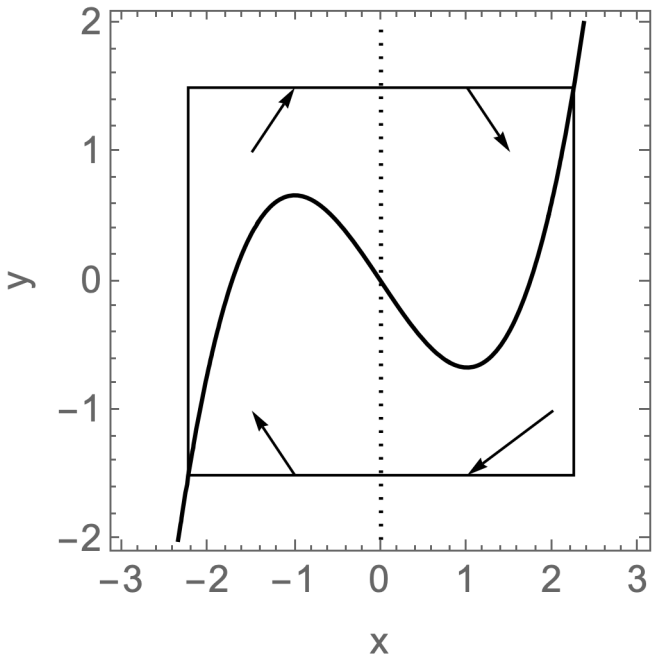
6. (trapping region for van der Pol) Consider the rectangular box shown in the figure below. The upper right and lower left corners co-incide with the cubic nullcline.



Sketch the nullclines and a vector representing the direction of the vector field in each of the four sectors created by the nullclines. Why isn't the box shown above a region that traps all trajectories? Where are trajectorieas able to escape the region?

Answers:

$\dot{y} > 0$ on part of the upper horizontal line, and $\dot{y} < 0$ on part of the lower horizontal line.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